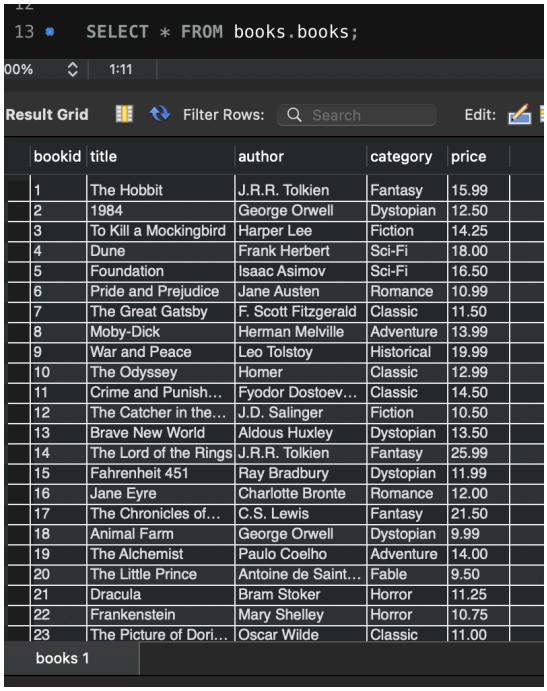
Luis Calvillo **Assignment 5**

Task 1

```sql
CREATE DATABASE books;
USE books;

CREATE TABLE books (
    bookid INT PRIMARY KEY,
    title VARCHAR(255),
    author VARCHAR(255),
    category VARCHAR(100),
    price DECIMAL(7,2)
);


SELECT * FROM books.books;
```



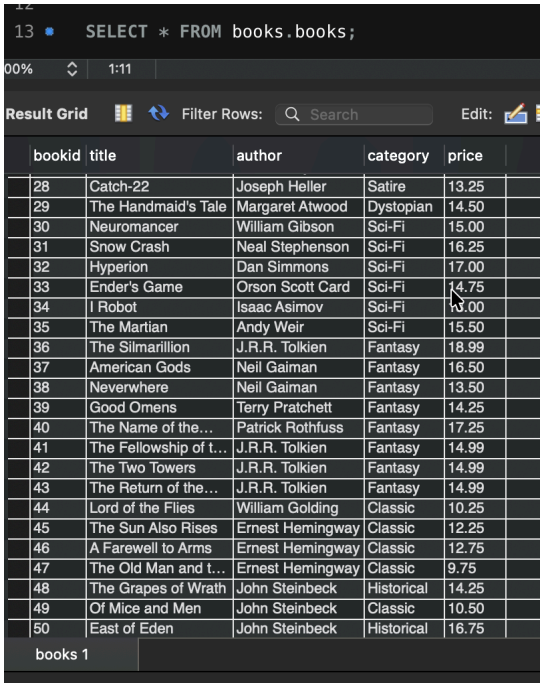
13 SELECT * FROM books.books;

100% 1:11

Result Grid Filter Rows: Search Edit:

bookid	title	author	category	price
1	The Hobbit	J.R.R. Tolkien	Fantasy	15.99
2	1984	George Orwell	Dystopian	12.50
3	To Kill a Mockingbird	Harper Lee	Fiction	14.25
4	Dune	Frank Herbert	Sci-Fi	18.00
5	Foundation	Isaac Asimov	Sci-Fi	16.50
6	Pride and Prejudice	Jane Austen	Romance	10.99
7	The Great Gatsby	F. Scott Fitzgerald	Classic	11.50
8	Moby-Dick	Herman Melville	Adventure	13.99
9	War and Peace	Leo Tolstoy	Historical	19.99
10	The Odyssey	Homer	Classic	12.99
11	Crime and Punish…	Fyodor Dostoev…	Classic	14.50
12	The Catcher in the…	J.D. Salinger	Fiction	10.50
13	Brave New World	Aldous Huxley	Dystopian	13.50
14	The Lord of the Rings	J.R.R. Tolkien	Fantasy	25.99
15	Fahrenheit 451	Ray Bradbury	Dystopian	11.99
16	Jane Eyre	Charlotte Bronte	Romance	12.00
17	The Chronicles of…	C.S. Lewis	Fantasy	21.50
18	Animal Farm	George Orwell	Dystopian	9.99
19	The Alchemist	Paulo Coelho	Adventure	14.00
20	The Little Prince	Antoine de Saint…	Fable	9.50
21	Dracula	Bram Stoker	Horror	11.25
22	Frankenstein	Mary Shelley	Horror	10.75
23	The Picture of Dori…	Oscar Wilde	Classic	11.00

books 1



13 SELECT * FROM books.books;

100% 1:11

Result Grid Filter Rows: Search Edit:

bookid	title	author	category	price
28	Catch-22	Joseph Heller	Satire	13.25
29	The Handmaid's Tale	Margaret Atwood	Dystopian	14.50
30	Neuromancer	William Gibson	Sci-Fi	15.00
31	Snow Crash	Neal Stephenson	Sci-Fi	16.25
32	Hyperion	Dan Simmons	Sci-Fi	17.00
33	Ender's Game	Orson Scott Card	Sci-Fi	14.75
34	I Robot	Isaac Asimov	Sci-Fi	13.00
35	The Martian	Andy Weir	Sci-Fi	15.50
36	The Silmarillion	J.R.R. Tolkien	Fantasy	18.99
37	American Gods	Neil Gaiman	Fantasy	16.50
38	Neverwhere	Neil Gaiman	Fantasy	13.50
39	Good Omens	Terry Pratchett	Fantasy	14.25
40	The Name of the…	Patrick Rothfuss	Fantasy	17.25
41	The Fellowship of t…	J.R.R. Tolkien	Fantasy	14.99
42	The Two Towers	J.R.R. Tolkien	Fantasy	14.99
43	The Return of the…	J.R.R. Tolkien	Fantasy	14.99
44	Lord of the Flies	William Golding	Classic	10.25
45	The Sun Also Rises	Ernest Hemingway	Classic	12.25
46	A Farewell to Arms	Ernest Hemingway	Classic	12.75
47	The Old Man and t…	Ernest Hemingway	Classic	9.75
48	The Grapes of Wrath	John Steinbeck	Historical	14.25
49	Of Mice and Men	John Steinbeck	Classic	10.50
50	East of Eden	John Steinbeck	Historical	16.75

books 1

Task 2

Express server.js

```js
const express = require('express');
const db = require('./mysql');
```

```
const pool = typeof db.promise === 'function' ? db.promise() : db;

const app = express();
const PORT = 3000;

function escapeHtml(value) {
   return String(value).replace(/[&<>"']/g, (char) => ({
       '&': '&amp;',
       '<': '&lt;',
       '>': '&gt;',
       '"': '&quot;',
       "'": '&#39;'
   })[char]);
}
// GET route to search books using URL query parameters.
app.get('/', async (req, res) => {
   const { category, author, format = 'json' } = req.query;

   if ([category, author, format].some(
       value => value !== undefined && typeof value !== 'string'
   )) {
       return res.status(400).json({ error: 'Invalid parameters' });
   }
   let sql = 'SELECT * FROM books WHERE 1=1';
   const values = [];

   // Filter by category.
   if (category) {
       sql += ' AND category = ?';
       values.push(category);
   }
   // Filter by author.
   if (author) {
       sql += ' AND author = ?';
       values.push(author);
   }

   try {
       const [books] = await pool.query(sql, values);
       // Return readable HTML when format=text.
       if (format.toLowerCase() === 'text') {
           let html = `
```

```
            <html>
            <head><title>Book Results</title></head>
            <body>
            <h1>Book Search Results</h1>
            <table border="1" cellpadding="10">
            <tr>
                <th>ID</th>
                <th>Title</th>
                <th>Author</th>
                <th>Category</th>
                <th>Price</th>
            </tr>
        `;
        books.forEach(book => {
            html += `
                <tr>
                    <td>${escapeHtml(book.bookid)}</td>
                    <td>${escapeHtml(book.title)}</td>
                    <td>${escapeHtml(book.author)}</td>
                    <td>${escapeHtml(book.category)}</td>
                    <td>$${escapeHtml(book.price)}</td>
                </tr>
            `;
        });

        html += '</table></body></html>';
        return res.send(html);
    }

    res.json(books);

  } catch (error) {
    console.error('Database error:', error);
    res.status(500).json({
        error: 'Failed to retrieve books'
    });
  }
});

// Starts server
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
```

```
});
```

Mysql connection

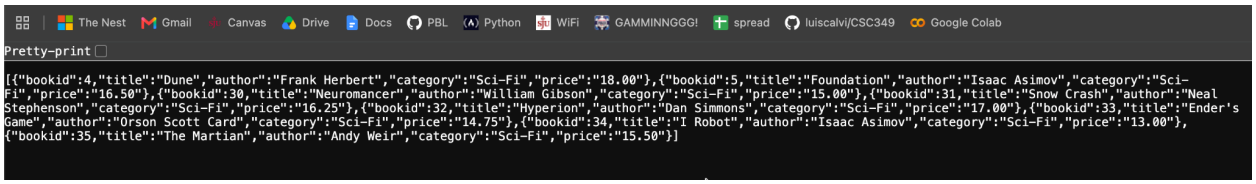
```
const mysql = require('mysql2/promise');

const pool = mysql.createPool({
   host: 'localhost',
   user: 'root',
   password: 'password',
   database: 'books',
   waitForConnections: true,
   connectionLimit: 10
});

module.exports = pool;
```

Task 3

http://localhost:3000/?category=Sci-Fi



http://localhost:3000/?author=George%20Orwell

